Compounding Impacts of Temperature and Invasive European Green Crabs on Hairy Shore Crabs

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Abstract

Invasive species and warming oceans drive ecosystem change, but their combined effects on native organisms are yet to be studied. The European green crab (*Carcinus maenas*) is an example of such. In this experiment we examined how elevated temperature (24°C) and exposure to European green crabs affect righting time, total movement, and respiration in hairy shore crabs compared to ambient conditions (14°C). We found that elevated temperature had a stronger effect than predator exposure, with shore crabs showing quicker righting times, increased activity, and higher mortality under warm conditions. Predator exposure did not produce any significant behavioral change, while respiration rates declined over time, suggesting acclimatization. These results indicate that ocean warming may be a critical driver of stress in hairy shore crabs, far greater than predator presence in shaping stress responses.

1. Introduction

A common disruptor of ecosystems worldwide is invasive species. Invasive species prey on native organisms but often have no predator to maintain an equilibrium in population, leading to unregulated population growth (Peller & Altermatt, 2024). Such organisms not only tip the scale of the ecosystem but also impact nutrient and chemical presence and distribution, which in turn can influence adjacent ecosystems and colonies as well. One example of an invasive species is the European green crab (*Carcinus maenas*).

European green crabs are an invasive species commonly found in intertidal habitats across the West Coast and are increasing in abundance in Puget Sound (Turner, 2025). Despite the removal of nearly 2 million individuals since 2022 their population continues to expand, including new detections in south central Puget Sound. Estuaries, eelgrass beds, shellfish habitats, and juvenile fish nursery areas are all considered to be under threat due to this expansion. It is important to note that the threats the different ecosystems face are not independent of each other; many organisms are threatened indirectly due to disrupted linkages in the food chain in a trophic cascade (Walsh et al., 2016). Damage at this scale requires significant resources to be reversed. However, damage estimates are arbitrary with predictions often underestimating the indirect effects, making it difficult to fully appreciate and respond to an influx of invasive species.

The hairy shore crab is a highly impacted species native to the Puget Sound region by European green crab presence both directly and indirectly; green crabs not only prey on hairy shore crabs but also reduce habitat and food availability due to common scavenging grounds and prey (Fisher et al., 2024). As a result, hairy shore crab populations may experience increased physiological stress including reduced growth, altered foraging behavior, and shifts in metabolic demand. These factors can lead to lower chances of survival and thus a lower likelihood of reproduction. To understand the degree at which such stress responses occur, metrics such as antipredator behavior and metabolic rate are utilized (Sih et al., 2009; Griffen et al., 2024). Sih and colleagues found that predatory stimuli result in behavioral changes in the prey (Sih et al., 2009). Griffen and colleagues found that shore crabs display increased respiration when exposed to sudden conditions with respiration decreasing after acclimatization (Griffen et al., 2024). But while studies on stress response of shore crabs exist, little is known about the stress response in relation to European green crab presence.

Warmer waters are positively correlated with European green crab abundance, providing evidence that global warming will benefit the reproductive success of said species (Yamada et al., 2021). Likewise, it is important to note that hairy shore crabs are also likely to be impacted by the new climate, potentially resulting in a change in how they respond to the presence of European green crabs. We do not know what that will be. In this study, we focused on the stress response of hairy shore crabs in response to the presence of European green crabs at differing temperature conditions. We measured travelled distance, righting time, and respiration prior to and after exposure to higher temperature and the European green crab to see if there were any behavioral or metabolic changes in response to each factor. We hypothesize that increased temperature and predator presence will contribute to more movement and delayed righting time, while displaying increased metabolism. While this research cannot identify the impact of European green crab presence on the ecosystem scale alongside climate change, it will establish an understanding of the stress factors in hairy shore crabs which can contribute to the conservation of the hairy shore crab population.

2. Methods

Two European green crabs and twelve hairy shore crabs were selected at random, while making sure that in the hairy shore crab group there are six females and six males. Males infected by parasites were excluded. Six shore crabs of each sex were randomly allocated to ambient and elevated treatment groups. Ambient conditions were set at 14°C and 35 ppt, while elevated conditions were set at 24°C and 35 ppt. For each group, two shore crabs, one male and another female, were randomly selected and labeled for distance travelled measurements. The other four from each group were labeled for respiration measurements. Each crab was patted with paper towel to dry and weighed.

For distance traveled, each crab was placed in a tank with an oyster mesh on one end and a glass wall on the other. A corridor of 32 cm length and 6 cm width was set up. Each shore crab was placed 2 cm from the oyster mesh and recorded for 30 seconds. Horizontal distance was measured after every stop of 2 or more seconds. Righting time was recorded by drying each shore crab, placing them on their back, and recording the time for the crab to right itself back up. Respiration was recorded through a resazurin assay. Each shore crab was placed in a well plate with 8 mL of resazurin. At 0, 15, 30, and 45 minutes after the shore crab was placed in resazurin, 200 µl of resazurin was sampled. After 45 minutes, the shore crabs were set to rest in an ambient condition tank before being returned to their treatment group tanks. All samples were analyzed in a spectrophotometer plate reader, and readings were normalized by weight.

The same day that the shore crabs were separated into groups, righting time, respiration, and distance traveled with no European green crab present were recorded. Then, all shore crabs were placed in their respective tanks for a week. Shore crabs were tested again for righting time, distance traveled, and respiration prior to exposure to the European green crab. The European green crab was then placed on the other side of the oyster mesh and distance traveled was recorded for all shore crabs. After exposure shore crabs were tested for righting time once more. The shore crabs were placed into their respective treatment temperatures for a week again and righting time prior to predator exposure was recorded. Each shore crab was exposed to the European green crab in the corridor tank for 30 seconds with the oyster mesh in-between, and respiration was recorded for the last time after exposure.

The mean and distribution of each variable for predator exposure and temperature were compared by visual estimates on the graph.

3. Results

Of the six crabs in the ambient group, all survived. However, three of six shore crabs in the elevated group died during the second week of treatment, with one dying after righting time for two-week exposure was measured. High volumes of floating matter were found in the elevated group's tank relative to the ambient temperature group. Two of three were crabs used for resazurin measurements; consequently, one non-resazurin crab was used for resazurin measurements instead.

Prior to treatment exposure, both treatment groups exhibited similar righting times, averaging around 0.80 seconds (Fig. 1A). The two also exhibited similar distances travelled, with two different behaviors; one constantly moving towards the glass wall edge, with another maintaining a relatively stationary position (Fig. 2A). A week after exposure to treatment conditions, shore crabs exposed to elevated conditions exhibited faster reaction speeds. While shore crabs in the ambient group maintained a relatively constant righting time, shore crabs in elevated conditions exhibited on average a 0.50 second shorter righting time (Fig. 1B). No change was observed between one and two weeks of exposure, although the variance in righting time did increase (Fig. 2C). A similar pattern was observed in travelled distance as both crabs in the elevated condition were the fastest to move towards the glass wall edge; on the contrary, ambient condition shore crabs were slower to make any movement and traveled significantly less (Fig. 2B). After exposure to predatory presence, the righting time and distance traveled between ambient and elevated treatment groups are nearly identical (Fig. 1D; Fig. 2C). Righting time after predatory presence in the ambient treatment group is similar to that of the ambient group when first sampled. The same is true for the elevated group, though righting time is higher compared to that of when they were exposed to warmer conditions for a week.

Prior to exposing the two groups to the treatments, resazurin readings show high respiration rates (Fig. 3). There is no difference in rate between ambient and elevated groups. Prior to exposure to the predator the rate of respiration decreased and stayed consistent after predatory exposure. There was also no difference in respiration rate between elevated and ambient temperature groups, though the ambient group in both cases had a bit more variability.

4. Discussion

Given that three of six shore crabs in the elevated condition exhibited death despite all other factors for the ambient and elevated group being equal, we concluded that elevated conditions are a contributing factor to stress in shore crabs. As shore crabs in elevated groups displayed faster and heightened activity, and as all shore crabs were fasted for the duration of the experiment, excessive energy usage is a likely cause of death. Significant amounts of debris were found in the water of the elevated group compared to the ambient group, which could be excrements released from the shore crabs during death.

Elevated temperature appeared to influence shore crab behavior at a greater scale than predation alone. Shore crabs exposed to warmer temperatures showed faster righting times and greater activity compared to those in ambient conditions. Previous studies have shown that warmer temperatures are positively associated with green crab abundance and reproductive success (Yamada et al., 2021; Nielsen et al., 2025). As such, elevated temperature may increase metabolic demand and behavioral activity in intertidal crab species on a broader scale as observed in hairy shore crabs. However, after predator exposure, behavioral differences between treatment groups became less distinct. Respiration rates also declined over time regardless of the treatment group. It is possible that elevated temperature initially increases activity in hairy shore crabs and, while prolonged exposure may have led to acclimatization, it did not necessarily reduce physiological stress to levels original to ambient conditions.

Predator exposure did not produce the behavioral changes originally hypothesized. Sih and colleagues found that prey often alter their behavior in response to predatory stimuli (Sih et al., 2009). In this study, however, shore crabs exposed to the European green crab displayed righting times and movements that became more similar between temperature groups. While one possible explanation is that shore crabs acclimated to predatory cues, this would mean the shore crabs all acclimated to the signal as soon as they were exposed. Another possibility is that the oyster mesh barrier acted as a safe sanctuary despite it being at close proximity to the predator. Rather than "running" from the predator, their first instinct may be to burrow and hide to minimize exposure, especially in a narrow channel with limited direction. And although European green crabs are known as predators and competitors of shore crabs (Fisher et al., 2024), the short exposure period used in this experiment may not have been sufficient to trigger stronger antipredator responses.

Respiration patterns also differed from what was hypothesized. Both treatment groups initially exhibited high respiration rates that later decreased and stabilized in two weeks. Griffen and colleagues observed similar patterns in shore crabs exposed to stressful conditions, with respiration increasing initially before declining after acclimatization (Griffen et al., 2024). Handling and treatment exposure may have the same influence, which in our design would be confounding factors as all crabs were handled prior to, during, and after measurements.

Overall, elevated temperature had a stronger effect on hairy shore crab behavior and survival than short-term exposure to invasive European green crabs. That said, it is worth noting the small sample size and significant variation that may obscure the data. Additionally, results were compared by visual comparison rather than statistical analyses. Despite these limitations, this study contributes to understanding how warming conditions and invasive European green crabs may influence native hairy shore crabs. By expanding this research to other native species, we can understand how invasive species alter ecosystems through trophic cascades and disrupted ecological interactions.

References

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Figures

Figure 1: Righting time of hairy shore crabs after A) sampling, B) a week of treatment exposure, C) two weeks of treatment exposure, and D) one week of treatment and predator exposure.

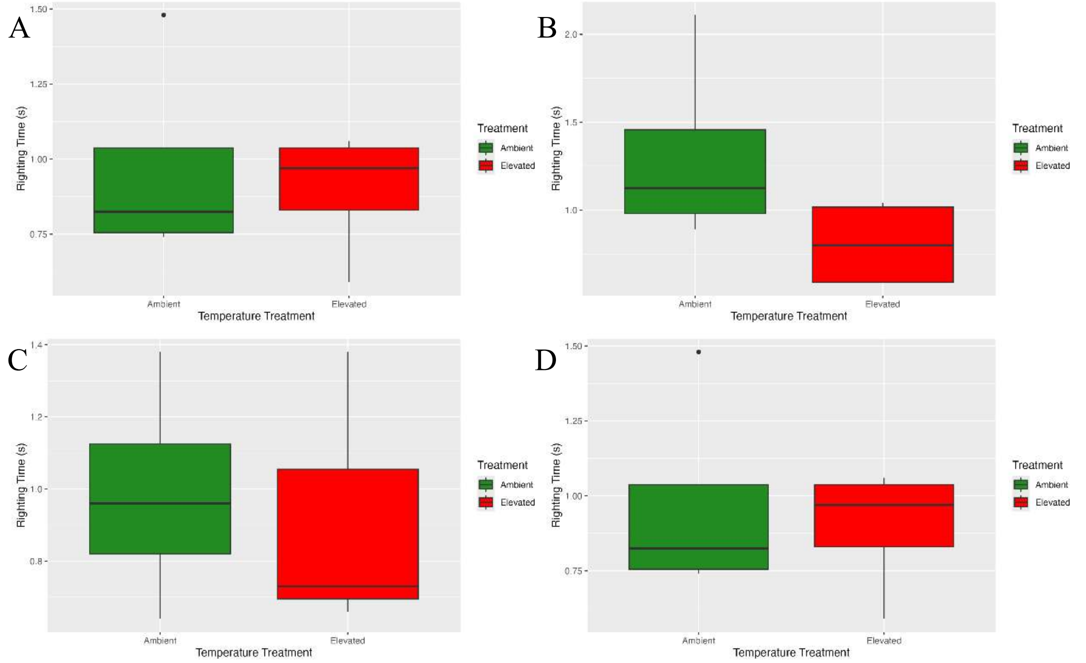


Figure 2: Horizontal distance traveled by shore crabs after A) sampling, B) one week of treatment exposure, and C) one week of treatment and predator exposure.

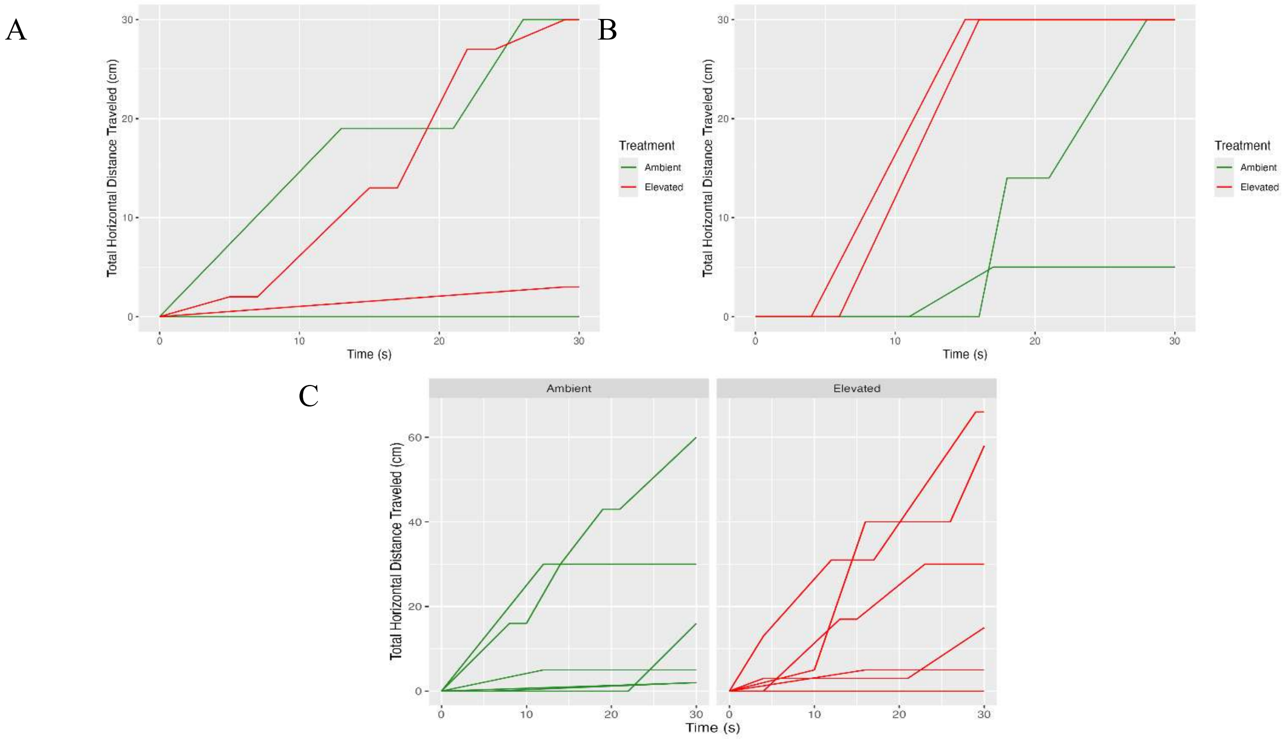
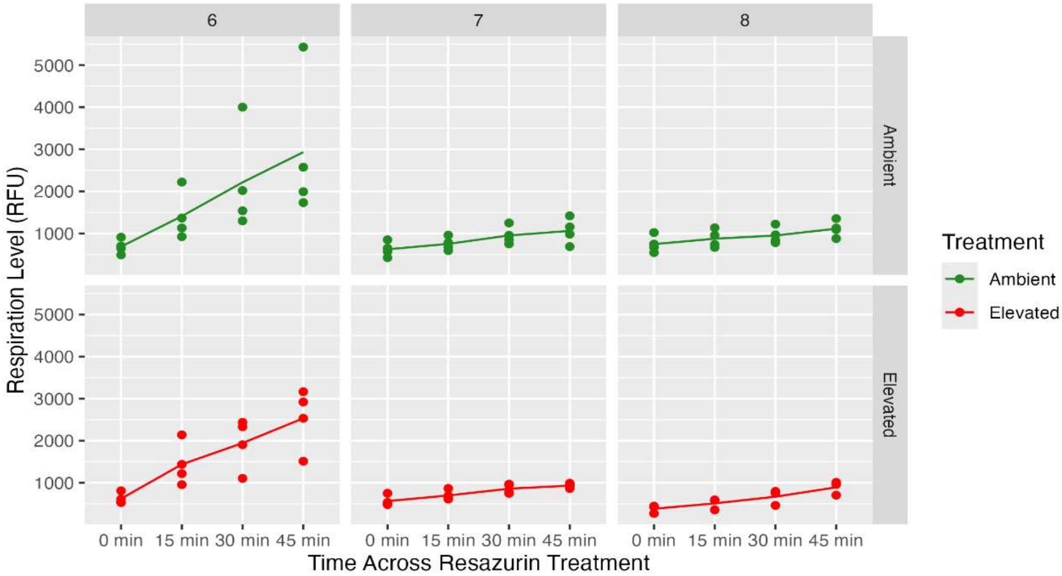


Figure 3: Resazurin assay readings by week 6, 7, and 8 in ambient and elevated temperatures. Week 6 represents the groups prior to treatment, week 7 indicates pre-predator exposure, and week 8 indicates post-predator exposure respiration.



6
7
8
Ambient
Elevated
5000
4000
3000
2000
1000
Respiration Level (RFU)
0 min 15 min 30 min 45 min
Time Across Resazurin Treatment
Treatment
Ambient
Elevated